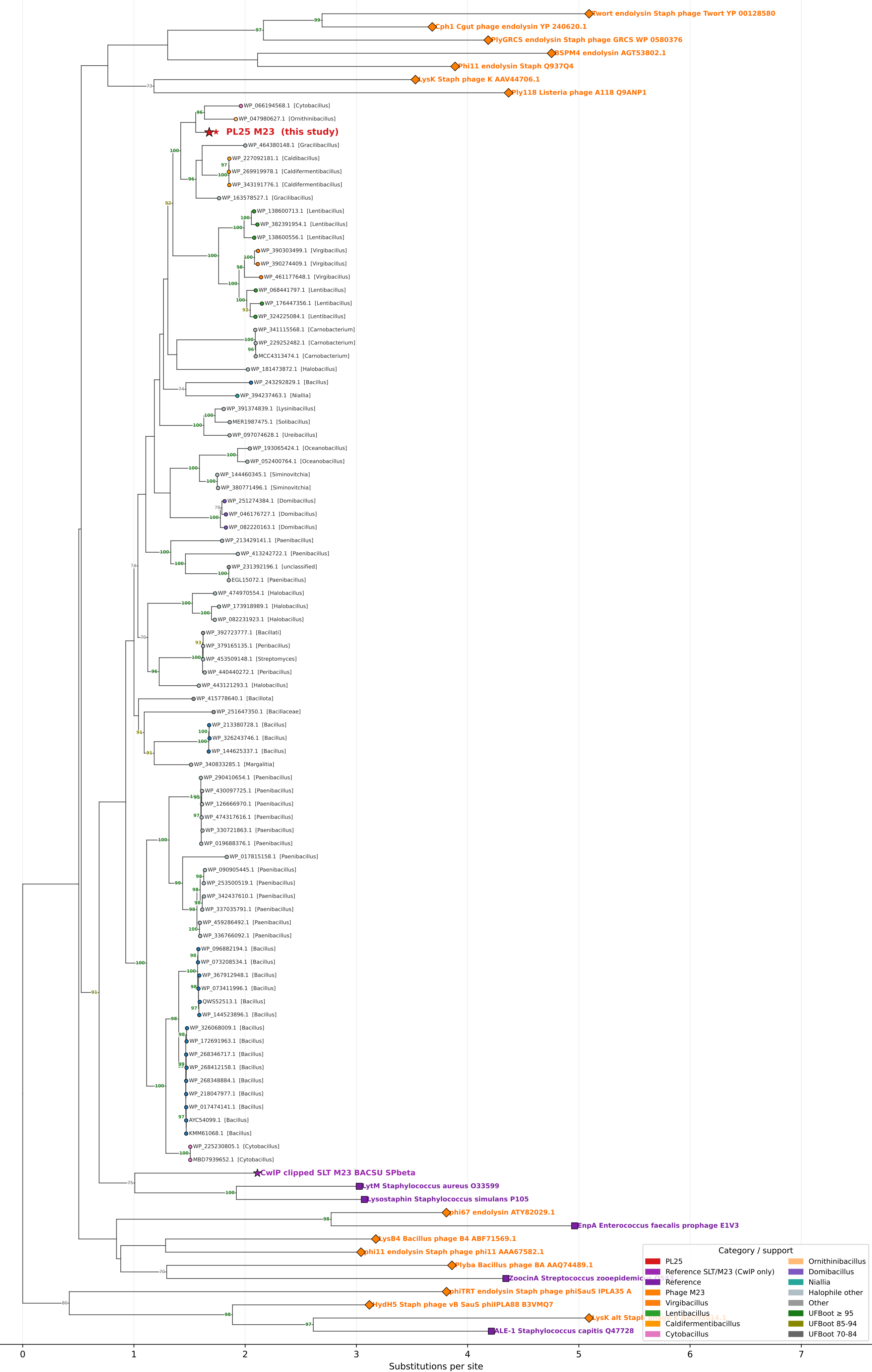




★ red = PL25 M23 (plasmid-encoded, this study) • ★ purple = CwIP-clipped (*B. subtilis* SP-β, only published SLT+M23 reference) • □ purple-dark = other M23 reference enzymes (lysostaphin, LytM, ALE-1, EnpA, zoocin A) • ◆ orange = phage-encoded M23 lysins • ● = halophile-Bacillaceae homologues from NCBI nr. CwIT is omitted from this M23 phylogeny because its C-terminal peptidase belongs to the NlpC/P60 cysteine-peptidase family (Pfam PF00877), not Pfam Peptidase_M23.